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RAPPORT ACADÉMIQUE

Partie Théorique et Partie Pratique

Customer Churn Prediction Using Machine Learning and Artificial Neural Networks

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Liste des abréviations:

IA / AI	Intelligence Artificielle / Artificial Intelligence
ML	Machine Learning
ANN	Artificial Neural Network (réseau de neurones artificiel)
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
SVM	Support Vector Machine
k-NN	k-Nearest Neighbors
CRM	Customer Relationship Management
TP / FP / TN / FN	True Positive / False Positive / True Negative / False Negative
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
PCA	Principal Component Analysis
GPU	Graphics Processing Unit
SHAP	SHapley Additive exPlanations
CRISP-DM	Cross-Industry Standard Process for Data Mining

Introduction Générale:

In today's digital economy, organizations operate in an increasingly competitive environment where customer satisfaction and customer retention have become fundamental determinants of long-term business success. The rapid evolution of information technology, digital services, and online platforms has significantly increased customer expectations while simultaneously providing consumers with numerous alternative products and services. Consequently, customers can easily switch from one provider to another whenever they perceive better value, lower prices, or higher service quality. This phenomenon, commonly known as customer churn, has become one of the most critical challenges facing modern organizations, particularly in highly competitive industries such as telecommunications, banking, insurance, e-commerce, healthcare, and subscription-based digital services [1]. Customer churn is defined as the process by which customers discontinue their relationship with a company by cancelling a subscription, terminating a service contract, or migrating to a competing provider. High churn rates negatively influence organizational profitability because acquiring new customers is considerably more expensive than retaining existing ones. Furthermore, customer attrition leads not only to direct revenue loss but also to increased marketing costs, reduced customer lifetime value, and weakened competitive positioning. Consequently, organizations continuously seek intelligent methods capable of identifying customers who are likely to churn before the actual departure occurs, enabling them to implement proactive retention strategies and personalized marketing campaigns [1], [2].

The exponential growth of digital data has transformed the manner in which organizations analyze customer behavior. Every interaction between customers and businesses—including purchases, online transactions, service usage, payment history, customer support activities, and demographic information—generates valuable data that can be exploited to support business decision-making. Traditional statistical methods often struggle to analyze these massive and complex datasets effectively. Therefore, Artificial Intelligence (AI), particularly Machine Learning (ML), has emerged as one of the most powerful technologies for extracting meaningful knowledge from data and producing highly accurate predictive

models [3].

Machine Learning enables computer systems to learn automatically from historical data without being explicitly programmed for every possible situation. Instead of relying on predefined rules, Machine Learning algorithms identify hidden relationships, recognize behavioral patterns, and continuously improve their predictive capabilities as more data become available. These characteristics have made Machine Learning one of the most influential technologies in predictive analytics and business intelligence. Today, Machine Learning supports decision-making in numerous domains, including fraud detection, medical diagnosis, recommendation systems, financial forecasting, cybersecurity, autonomous vehicles, and customer behavior prediction [3], [4].

Among the numerous applications of Machine Learning, Customer Churn Prediction has received considerable attention from both researchers and industrial practitioners. Customer churn prediction aims to estimate the probability that a customer will terminate their relationship with a company based on historical behavioral and demographic information. Accurate churn prediction enables organizations to identify high-risk customers, optimize marketing resources, reduce customer attrition, increase customer loyalty, and maximize long-term profitability. Recent systematic reviews demonstrate that Machine Learning techniques, particularly ensemble methods and Deep Learning models, have significantly improved churn prediction performance across multiple industrial sectors [1].

Recent advances in Deep Learning have further enhanced predictive modeling capabilities through the development of Artificial Neural Networks (ANNs). Inspired by the biological structure of the human brain, Neural Networks consist of interconnected computational units capable of learning highly complex nonlinear relationships from large datasets. Compared with many traditional Machine Learning algorithms, Neural Networks often achieve superior predictive performance when sufficient training data are available. Consequently, they have become one of the most widely adopted techniques for solving complex classification and prediction problems, including customer churn prediction [4].

The primary objective of this report is to investigate the application of Machine Learning and Artificial Neural Networks in Customer Churn Prediction. The first part presents the theoretical foundations of Machine Learning and Neural Networks, including their concepts, algorithms, learning mechanisms, advantages, limitations, and real-world applications. The second part focuses on the practical implementation of customer churn prediction using Python within the Google Colab environment. Two predictive approaches will be developed and evaluated: a traditional Machine Learning model and an Artificial Neural Network model. Their predictive performances will then be compared using standard evaluation metrics to determine the most appropriate technique for accurately predicting customer churn.

Ultimately, this report demonstrates how modern predictive analytics enables organizations to transform raw customer data into valuable business knowledge, facilitating intelligent decision-making, improving customer retention strategies, reducing financial losses, and strengthening long-term customer relationships in today's highly competitive digital economy.

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PARTIE I — CADRE THÉORIQUE

Chapitre 1 : Le Machine Learning, fondements et application à la prédiction du churn client:

1.1 Introduction to Machine Learning:

Machine Learning (ML) has become one of the fastest-growing fields within Artificial Intelligence (AI) and data science. The continuous increase in digital data generated through online services, mobile applications, social media, sensors, and business transactions has created an urgent need for intelligent systems capable of automatically analyzing massive datasets and extracting meaningful knowledge. Traditional programming approaches, where every rule must be explicitly defined by developers, are no longer sufficient to solve complex real-world problems. Machine Learning addresses this limitation by enabling computers to learn directly from data and improve their performance through experience without requiring explicit programming for every possible situation [1].

The concept of Machine Learning was first introduced by Arthur Samuel in 1959, who described it as the ability of computers to learn without being explicitly programmed. Later, Tom M. Mitchell provided one of the most widely accepted academic definitions, stating that “A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T, as measured by P, improves with experience E” [2]. This definition highlights the three fundamental components of every Machine Learning system: experience (training data), task (prediction or classification), and performance measure (evaluation metrics).

Unlike traditional software systems that rely on predefined instructions, Machine Learning algorithms automatically discover hidden patterns, correlations, and relationships from historical data. These learned patterns are then used to make predictions, classify observations, detect anomalies, or support intelligent decision-making. As more data become available, Machine Learning models can continuously improve their predictive performance, making them highly adaptable to dynamic environments [2].

Today, Machine Learning has become an indispensable technology across numerous industries. In healthcare, it supports disease diagnosis and medical image analysis; in finance, it is widely used for fraud detection and credit risk assessment; in manufacturing, it enables predictive maintenance; while in e-commerce and digital marketing, it powers recommendation systems and customer behavior analysis. One of its most valuable business applications is Customer Churn Prediction, where Machine Learning models analyze customer demographics, service usage, transaction history, and behavioral characteristics to identify customers who are likely to leave a company. These predictions enable organizations to implement targeted retention strategies, reduce customer attrition, and improve long-term profitability [3].

The remarkable success of Machine Learning has been driven by three major factors: the exponential growth of digital data, significant improvements in computing power, and

continuous advances in learning algorithms. Together, these factors have transformed Machine Learning from a purely academic discipline into a fundamental technology supporting intelligent systems and data-driven decision-making worldwide. Consequently, understanding the principles, methodologies, and applications of Machine Learning is essential before exploring more advanced techniques such as Artificial Neural Networks, which constitute the next stage of this report.

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1.2 History and Evolution of Machine Learning:

The history of Machine Learning dates back to the early development of Artificial Intelligence during the mid-twentieth century. The idea of creating machines capable of learning and making intelligent decisions fascinated researchers long before the availability of modern computers. One of the earliest milestones was the work of Alan Turing, who proposed that machines could simulate intelligent behavior through learning and adaptation. In his famous paper Computing Machinery and Intelligence (1950), Turing introduced the concept that a machine could demonstrate intelligence if its behavior became indistinguishable from that of a human, a concept later known as the Turing Test

A significant breakthrough occurred in 1959 when Arthur Samuel introduced the term Machine Learning while developing a computer program capable of playing the game of checkers. Samuel defined Machine Learning as “the field of study that gives computers the ability to learn without being explicitly programmed.” His work demonstrated that computers could improve their performance over time by learning from experience rather than relying solely on manually written rules. This achievement marked the beginning of Machine Learning as an independent scientific discipline [2].

During the 1960s and 1970s, researchers developed several statistical learning methods and pattern recognition techniques. Although computational resources were limited, these studies established the mathematical foundations of supervised learning, probability theory, optimization, and classification algorithms. The development of algorithms such as the Perceptron by Frank Rosenblatt represented one of the earliest attempts to create artificial neural models capable of learning from data [3].

The 1980s witnessed renewed interest in Artificial Neural Networks following the introduction of the backpropagation algorithm by Rumelhart, Hinton, and Williams. This algorithm enabled multi-layer neural networks to learn efficiently by adjusting their internal weights during training. At the same time, Decision Trees, Bayesian Networks, and other statistical learning techniques became increasingly popular for solving classification and prediction problems [4].

Machine Learning experienced remarkable growth during the 1990s with the publication of Tom M. Mitchell’s book Machine Learning, which became one of the most influential academic references in the field. During this period, researchers introduced several powerful algorithms, including Support Vector Machines (SVM), Ensemble Learning methods, and improved Decision Tree algorithms. The increasing availability of digital datasets also

encouraged the practical application of Machine Learning in business, medicine, finance, and engineering [5].

The beginning of the twenty-first century marked the era of Big Data, cloud computing, and high-performance graphics processing units (GPUs). These technological advances enabled Machine Learning algorithms to process millions of observations efficiently and accelerated the development of Deep Learning. Artificial Neural Networks evolved into Deep Neural Networks with multiple hidden layers capable of learning highly complex patterns from enormous datasets. This progress revolutionized fields such as computer vision, speech recognition, natural language processing, autonomous vehicles, and predictive analytics [6].

Today, Machine Learning has become an essential component of modern Artificial Intelligence. Organizations across various industries rely on Machine Learning to support decision-making, automate complex processes, improve operational efficiency, and deliver personalized customer experiences. One of the most important business applications is Customer Churn Prediction, where Machine Learning algorithms analyze customer behavior to estimate the probability of customer attrition. Recent advances in Deep Learning and Artificial Neural Networks have further improved prediction accuracy, making them indispensable tools for customer relationship management and business intelligence [7].

The continuous evolution of Machine Learning is driven by three major factors: the exponential growth of digital data, improvements in computational power, and the development of more sophisticated learning algorithms. As these technologies continue to advance, Machine Learning is expected to play an even greater role in solving complex real-world problems and supporting intelligent decision-making across numerous scientific and industrial domains.

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1.3 Types of Machine Learning:

Machine Learning algorithms are generally classified into three main categories according to the learning strategy adopted during model training: Supervised Learning, Unsupervised Learning, and Reinforcement Learning. Each category addresses different types of problems and utilizes distinct learning mechanisms depending on the availability of labeled data and the desired objective [1].

1.3.1 Supervised Learning:

Supervised Learning is the most widely used paradigm in Machine Learning. In this approach, the algorithm is trained using a labeled dataset, where each observation consists of input variables (features) and a corresponding target variable (label). The objective of the learning process is to discover the relationship between inputs and outputs so that the trained model can accurately predict the target value for previously unseen data [2].

Supervised Learning is mainly applied to two categories of problems: classification and regression. Classification aims to predict discrete classes, such as determining whether a customer will churn or remain loyal. Regression, on the other hand, predicts continuous numerical values, such as estimating future sales or customer lifetime value. Several algorithms have been successfully applied to supervised learning tasks, including Logistic Regression, Decision Trees, Random Forests, Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), Naïve Bayes, Gradient Boosting, and Artificial Neural Networks [3].

In the context of Customer Churn Prediction, Supervised Learning is considered the most appropriate approach because historical customer datasets contain labeled information indicating whether each customer has churned or not. By learning from these historical records, predictive models can estimate the probability that a current customer will leave the company, enabling organizations to implement proactive retention strategies and improve customer relationship management [4].

1.3.2 Unsupervised Learning :

Unlike Supervised Learning, Unsupervised Learning deals with datasets that do not contain predefined output labels. The objective is to identify hidden structures, similarities, or patterns within the data without prior knowledge of the expected results. This learning paradigm is particularly useful for exploratory data analysis and knowledge discovery [2].

The most common application of Unsupervised Learning is clustering, where observations with similar characteristics are grouped into clusters. Popular clustering algorithms include K-Means, Hierarchical Clustering, and DBSCAN. Another important application is dimensionality reduction, where techniques such as Principal Component Analysis (PCA) reduce the number of variables while preserving the essential information contained in the dataset [5].

In business analytics, Unsupervised Learning enables organizations to segment customers according to purchasing behavior, demographic characteristics, and service usage patterns. These customer segments support personalized marketing strategies, targeted advertising, and improved customer relationship management [4].

1.3.3 Reinforcement Learning:

Reinforcement Learning represents a different learning paradigm in which an intelligent agent interacts with an environment by performing actions and receiving feedback in the form of rewards or penalties. Instead of learning from labeled examples, the agent continuously improves its decision-making strategy by maximizing cumulative rewards over time [6].

Reinforcement Learning has achieved remarkable success in robotics, autonomous vehicles,

industrial automation, recommendation systems, and game-playing applications. Although it is less frequently applied to Customer Churn Prediction than Supervised Learning, it can support dynamic customer engagement by continuously adapting marketing actions based on customer responses and optimizing long-term retention strategies [6].

Comparison of Machine Learning Types:

Each Machine Learning paradigm has unique strengths and is designed for specific types of problems. Supervised Learning is the preferred approach when labeled datasets are available, particularly for prediction and classification tasks such as Customer Churn Prediction. Unsupervised Learning is primarily used for discovering hidden structures and customer segmentation, while Reinforcement Learning focuses on sequential decision-making through interaction with an environment. Selecting the appropriate learning paradigm depends on the nature of the available data, business objectives, and the complexity of the problem being addressed [1], [4].

Table 1.1. Comparison of Machine Learning Types :

Supervised Learning	Labeled Data	Classification and Regression	Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), k-Nearest Neighbors (k-NN), Artificial Neural Networks	Customer Churn Prediction, Fraud Detection, Disease Diagnosis, Spam Detection
Unsupervised Learning	Unlabeled Data	Pattern Discovery and Clustering	K-Means, Hierarchical Clustering, DBSCAN, Principal Component Analysis (PCA)	Customer Segmentation, Market Basket Analysis, Anomaly Detection
Reinforcement Learning	Rewards and Penalties	Learning Optimal Actions through Interaction	Q-Learning, SARSA, Deep Q-Network (DQN), Policy Gradient	Robotics, Autonomous Vehicles, Game Playing, Recommendation Systems

Source: Adapted from Mitchell (1997), Goodfellow et al. (2016), and Manzoor et al. (2024).

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1.4 Machine Learning Workflow:

Developing an effective Machine Learning model requires following a systematic workflow that transforms raw data into reliable predictive models. Although the implementation may vary depending on the problem and the selected algorithm, most Machine Learning projects follow a sequence of well-defined stages. These stages ensure that the developed model is accurate, reliable, and capable of generalizing to new data. A structured workflow also helps researchers identify potential issues during model development and improve the overall performance of the predictive system [1], [2].

1.4.1 Problem Definition :

The first stage of every Machine Learning project is defining the problem to be solved. A clear understanding of the business objective helps determine the type of Machine Learning task, the required data, and the appropriate evaluation metrics. In the case of Customer Churn Prediction, the objective is to predict whether a customer is likely to leave a company based on historical customer information. Clearly defining the problem ensures that the selected Machine Learning techniques are aligned with the organization’s business goals [1].

1.4.2 Data Collection:

The second stage consists of collecting relevant data from different sources. Data quality plays a crucial role in the success of any Machine Learning project because predictive models learn directly from the available information. Customer datasets may include demographic characteristics, service usage records, billing information, payment history, customer support interactions, and previous churn behavior. These data are generally obtained from Customer Relationship Management (CRM) systems, transactional databases, or online platforms [3].

1.4.3 Data Preprocessing:

Real-world datasets usually contain missing values, duplicated observations, inconsistent formats, and noisy information. Therefore, preprocessing is an essential step before model training. This stage includes handling missing values, removing duplicate records, correcting inconsistencies, encoding categorical variables, normalizing numerical features, and detecting outliers. Proper preprocessing significantly improves model accuracy and reduces the risk of biased predictions [2].

1.4.4 Feature Engineering:

Feature engineering involves selecting, transforming, or creating variables that improve the predictive performance of Machine Learning models. Well-designed features provide more meaningful information to learning algorithms and increase model accuracy. For Customer Churn Prediction, useful features may include customer tenure, monthly charges, total charges, contract type, payment method, internet service type, and customer support history. Feature selection techniques also help eliminate irrelevant variables and reduce computational complexity [3].

1.4.5 Model Training:

After preparing the dataset, the selected Machine Learning algorithm is trained using historical customer data. During training, the algorithm learns the relationships between the input variables and the target variable (customer churn). Various supervised learning algorithms can be employed, including Logistic Regression, Decision Trees, Random Forests, Support Vector Machines, Gradient Boosting, and Artificial Neural Networks. The selection of the most appropriate algorithm depends on dataset characteristics and predictive objectives [1].

1.4.6 Model Evaluation:

Once the training process is completed, the predictive model must be evaluated using previously unseen data. Model evaluation measures how well the trained model generalizes to new observations. Several performance metrics are commonly used in classification problems, including Accuracy, Precision, Recall, F1-score, Receiver Operating Characteristic (ROC) Curve, Area Under the Curve (AUC), and the Confusion Matrix. These metrics provide valuable insights into the strengths and weaknesses of predictive models and support the selection of the best-performing algorithm [4].

1.4.7 Model Deployment:

The final stage consists of deploying the trained Machine Learning model into a real business environment. In Customer Churn Prediction, the deployed model continuously analyzes customer information and estimates the probability of customer attrition. Business managers can then use these predictions to launch personalized retention campaigns, improve customer satisfaction, and optimize marketing investments. Continuous monitoring and periodic model retraining are also necessary to maintain prediction accuracy as customer behavior changes over time [2].

Following a structured Machine Learning workflow improves the reliability and effectiveness of predictive models while ensuring that the developed solution satisfies business objectives. This workflow will also be adopted in the practical part of this report, where Customer Churn Prediction models will be implemented and evaluated using Python in Google Colab.

Figure 1.2 – Machine Learning Workflow

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1.5 Machine Learning Algorithms for Customer Churn Prediction:

Machine Learning offers a wide range of algorithms for solving classification and prediction problems. Selecting the appropriate algorithm is one of the most important decisions during the development of a predictive model because different algorithms produce different levels of accuracy, computational complexity, interpretability, and robustness. In Customer Churn Prediction, the objective is to classify customers into two categories: customers who are likely to churn and customers who are expected to remain with the company. Since this problem represents a binary classification task, several supervised Machine Learning algorithms have been successfully applied in both academic research and industrial

applications [1].

1.5.1 Logistic Regression :

Logistic Regression is one of the simplest and most widely used supervised Machine Learning algorithms for binary classification problems. Unlike linear regression, which predicts continuous numerical values, Logistic Regression estimates the probability that an observation belongs to a particular class using the logistic (sigmoid) function. The predicted probability ranges between 0 and 1, allowing the model to classify customers as churners or non-churners based on a predefined threshold.

Despite its simplicity, Logistic Regression remains a powerful baseline model for Customer Churn Prediction because it is computationally efficient, easy to interpret, and performs well when the relationship between variables is approximately linear. Moreover, the model provides coefficients that help analysts understand how each customer attribute influences the probability of churn [2].

1.5.2 Decision Tree :

Decision Trees are non-parametric supervised learning algorithms that classify observations through a sequence of hierarchical decision rules. Each internal node represents a decision based on one feature, each branch corresponds to a possible outcome of that decision, and each terminal node represents the final predicted class.

Decision Trees are particularly attractive because they produce highly interpretable models that resemble human decision-making processes. In Customer Churn Prediction, a Decision Tree may first divide customers according to contract type, then monthly charges, followed by customer tenure, until a final prediction is obtained. However, individual Decision Trees are sensitive to noisy data and may suffer from overfitting if the tree becomes excessively deep [3].

1.5.3 Random Forest:

Random Forest is an ensemble learning algorithm that combines multiple Decision Trees to improve prediction accuracy and reduce overfitting. Instead of relying on a single tree, Random Forest constructs hundreds of Decision Trees using randomly selected subsets of training data and predictor variables. The final prediction is determined through majority voting among all individual trees.

Random Forest is widely recognized as one of the most effective algorithms for Customer Churn Prediction because of its high predictive performance, robustness against noisy data, and ability to model complex nonlinear relationships. Furthermore, it provides estimates of feature importance, allowing organizations to identify the variables that most strongly influence customer churn [3].

1.5.4 Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised learning algorithm designed to separate observations by identifying the optimal decision boundary, known as the hyperplane, which maximizes the distance between different classes. When the data are not linearly separable, SVM employs kernel functions to transform the original feature space into a higher-dimensional space where linear separation becomes possible.

SVM performs particularly well when dealing with high-dimensional datasets and complex classification problems. Nevertheless, its computational cost increases significantly for very large datasets, which may limit its practical use in industrial-scale Customer Churn Prediction systems [4].

1.5.5 k-Nearest Neighbors (k-NN) :

The k-Nearest Neighbors algorithm is an instance-based learning method that classifies new observations according to the classes of their nearest neighboring observations within the training dataset. The predicted class is determined by majority voting among the k closest neighbors measured using distance metrics such as Euclidean distance.

Although k-NN is simple to implement and often produces satisfactory results for small

datasets, its computational complexity becomes considerably higher as the size of the dataset increases. Consequently, it is less frequently adopted in large-scale customer churn prediction applications [2].

1.5.6 Artificial Neural Networks (ANN):

Artificial Neural Networks represent one of the most advanced Machine Learning techniques for classification and prediction. Inspired by the biological nervous system, Neural Networks consist of interconnected artificial neurons organized into input, hidden, and output layers. Through an iterative learning process known as backpropagation, the network automatically adjusts its internal parameters to minimize prediction errors. Artificial Neural Networks have demonstrated remarkable performance in Customer Churn Prediction because they can learn highly complex nonlinear relationships between customer characteristics and churn behavior. When sufficient training data are available, Neural Networks often outperform traditional Machine Learning algorithms in terms of predictive accuracy. Consequently, they have become increasingly popular in customer analytics and predictive business intelligence [5].

1.5.7 Comparison of Algorithms:

Each Machine Learning algorithm possesses unique strengths and limitations. Logistic Regression provides excellent interpretability, Decision Trees facilitate transparent decision-making, Random Forest offers high predictive accuracy and robustness, SVM performs well for complex classification tasks, k-NN is simple yet computationally expensive for large datasets, while Artificial Neural Networks excel at modeling highly nonlinear relationships. Therefore, selecting the most appropriate algorithm depends on factors such as dataset size, computational resources, interpretability requirements, and predictive objectives.

For Customer Churn Prediction, numerous comparative studies have concluded that ensemble methods such as Random Forest and advanced Deep Learning models based on Artificial Neural Networks generally achieve the highest predictive performance, particularly when trained on large customer datasets containing diverse behavioral and demographic features [1], [5].

Source: Adapted from Mitchell (1997), Bishop (2006), Goodfellow et al. (2016), and Manzoor et al. (2024).

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* Manzoor, A., Qureshi, M. A., Kidney, E., & Longo, L. (2024). A Review on Machine Learning Methods for Customer Churn Prediction and Recommendations for Business Practitioners. IEEE Access.

Figure 1.3 – Popular Machine Learning Algorithms for Customer Churn Prediction

[1] Manzoor, A., Qureshi, M. A., Kidney, E., & Longo, L. (2024). A Review on Machine Learning Methods for Customer Churn Prediction and Recommendations for Business Practitioners. IEEE Access, 12, 70434–70463.[1] Manzoor, A., Qureshi, M. A., Kidney, E., & Longo, L. (2024). A Review on Machine Learning Methods for Customer Churn Prediction and Recommendations for Business Practitioners. IEEE Access, 12, 70434–70463.

[2] Mitchell, T. M. (1997). Machine Learning. McGraw-Hill.[2] Mitchell, T. M. (1997). Machine Learning. McGraw-Hill.

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1.6 Performance Evaluation Metrics:

After developing and training a Machine Learning model, it is essential to evaluate its predictive performance before deploying it in real-world applications. Model evaluation determines how accurately the trained model predicts unseen data and whether it can generalize beyond the training dataset. In classification problems such as Customer Churn Prediction, relying on a single metric is often insufficient because datasets are frequently imbalanced, meaning that the number of customers who churn is much smaller than those who remain. Therefore, several evaluation metrics are used to provide a comprehensive assessment of the model's performance, including the Confusion Matrix, Accuracy, Precision, Recall, F1-Score, Receiver Operating Characteristic (ROC) Curve, and Area Under the Curve (AUC). These metrics help researchers and practitioners compare different Machine Learning algorithms and select the most suitable model for deployment [1], [2].

1.6.1 Confusion Matrix:

The Confusion Matrix is one of the most fundamental tools for evaluating classification models. It summarizes prediction results by comparing the predicted classes with the actual classes. For binary classification, the confusion matrix consists of four possible outcomes:

* True Positive (TP): Customers correctly predicted as churners.
* True Negative (TN): Customers correctly predicted as non-churners.
* False Positive (FP): Customers incorrectly predicted as churners although they remain.
* False Negative (FN): Customers incorrectly predicted as non-churners although they actually churn.

The confusion matrix serves as the foundation for calculating most classification evaluation metrics and provides valuable insights into the types of prediction errors made by the model.

Table 1.3. Confusion Matrix

Churn		True Positive (TP)		False Negative (FN)	
No Churn		False Positive (FP)		True Negative (TN)	

Source: Adapted from Bishop (2006) and NCBI Bookshelf – Evaluating Machine Learning Models and Their Diagnostic Value

1.6.2 Accuracy:

Accuracy measures the proportion of correctly classified observations among all observations in the dataset. It is one of the simplest and most commonly used evaluation metrics. However, when the dataset is highly imbalanced, accuracy alone may provide a misleading indication of model performance because a classifier may achieve a high accuracy while failing to identify minority-class observations correctly. Therefore, additional metrics such as Precision and Recall should also be conside

1.6.3 Precision:

Precision measures the proportion of correctly predicted positive observations among all observations predicted as positive. High precision indicates that when the model predicts customer churn, the prediction is usually correct. Precision is particularly important in applications where false positive predictions are costly.

1.6.4 Recall :

Recall, also known as Sensitivity or True Positive Rate, measures the proportion of actual positive observations that are correctly identified by the model. In Customer Churn

Prediction, Recall indicates how effectively the model identifies customers who are truly at risk of leaving the company. High Recall is desirable when missing churning customers would have significant business consequences.

1.6.5 F1-Score:

The F1-Score combines Precision and Recall into a single performance metric by calculating their harmonic mean. It is particularly useful when evaluating classification models on imbalanced datasets because it balances the trade-off between false positives and false negatives. A higher F1-Score indicates a better balance between Precision and Recall.

1.6.6 ROC Curve and AUC:

The Receiver Operating Characteristic (ROC) Curve is a graphical evaluation method that illustrates the relationship between the True Positive Rate (Recall) and the False Positive Rate (FPR) across different classification thresholds. It provides a comprehensive visualization of the classifier's ability to distinguish between positive and negative classes. The Area Under the ROC Curve (AUC) summarizes the ROC curve into a single numerical value ranging from 0 to 1. An AUC value close to 1.0 indicates excellent classification performance, whereas a value close to 0.5 suggests that the model performs no better than random guessing. In Customer Churn Prediction, ROC-AUC is widely used to compare different predictive models because it is less sensitive to class imbalance than accuracy alone.

Table 1.4. Summary of Classification Evaluation Metrics

Accuracy	1.0	Measures the overall proportion of correct predictions
Precision	1.0	Measures the correctness of positive predictions
Recall (Sensitivity)	1.0	Measures the ability to detect actual positive cases
F1-Score	1.0	Balances Precision and Recall
ROC-AUC	Close to 1	Measures the model's discrimination ability

Adapted from Bishop (2006), Goodfellow et al. (2016), and recent reviews on machine learning evaluation metrics.

1] C. M. Bishop, Pattern Recognition and Machine Learning. Springer, 2006.1] C. M. Bishop, Pattern Recognition and Machine Learning. Springer, 2006.

[2] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016.[2] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016.

[3] T. M. Mitchell, Machine Learning. McGraw-Hill, 1997.[3] T. M. Mitchell, Machine Learning. McGraw-Hill, 1997.

[4] A. Manzoor, M. A. Qureshi, E. Kidney, and L. Longo, "A Review on Machine Learning Methods for Customer Churn Prediction and Recommendations for Business Practitioners," IEEE Access, vol. 12, 2024.[4] A. Manzoor, M. A. Qureshi, E. Kidney, and L. Longo, "A Review on Machine Learning Methods for Customer Churn Prediction and Recommendations for Business Practitioners," IEEE Access, vol. 12, 2024.

1.7 Applications of Machine Learning :

Machine Learning has become one of the most influential technologies in the modern digital era due to its ability to extract meaningful patterns from large volumes of data and support intelligent decision-making. Advances in computational power, cloud computing, and big data technologies have significantly expanded the range of Machine Learning applications across various sectors. Today, Machine Learning is used in healthcare, finance,

cybersecurity, manufacturing, transportation, education, agriculture, and business analytics to automate processes, improve operational efficiency, and generate accurate predictions [1], [2].

One of the most significant application areas of Machine Learning is the healthcare sector. Machine Learning algorithms assist healthcare professionals in diagnosing diseases, analyzing medical images, predicting disease progression, and recommending personalized treatment plans. Deep Learning models, particularly Convolutional Neural Networks (CNNs), have achieved remarkable success in medical image analysis, including the detection of cancer, diabetic retinopathy, and cardiovascular diseases [2].

In the financial sector, Machine Learning plays a crucial role in fraud detection, credit scoring, algorithmic trading, and financial risk assessment. Banks and financial institutions analyze millions of daily transactions using Machine Learning algorithms to identify suspicious activities in real time and reduce financial losses. Predictive models also support loan approval decisions by estimating the creditworthiness of customers based on historical financial data [3].

Machine Learning is also extensively applied in cybersecurity, where intelligent algorithms continuously monitor network traffic, detect malicious activities, and identify cyberattacks before they compromise information systems. Anomaly detection techniques enable organizations to recognize unusual user behavior, malware infections, and unauthorized access attempts, thereby strengthening information security [4].

Another important application is found in the manufacturing industry, where Machine Learning supports predictive maintenance. By analyzing data collected from industrial sensors, predictive models estimate when equipment is likely to fail, allowing maintenance to be performed before unexpected breakdowns occur. This approach minimizes downtime, reduces maintenance costs, and increases production efficiency [2].

The transportation sector has also benefited significantly from Machine Learning technologies. Autonomous vehicles rely on Machine Learning and Deep Learning algorithms to recognize traffic signs, detect pedestrians, avoid obstacles, and make real-time driving decisions. Additionally, logistics companies use predictive analytics to optimize delivery routes, reduce fuel consumption, and improve supply chain management [2].

Machine Learning has transformed the field of e-commerce and digital marketing. Online platforms analyze customer browsing history, purchasing behavior, and product preferences to recommend personalized products and services. Recommendation systems employed by companies such as Amazon and Netflix improve customer satisfaction by suggesting products or content that match individual preferences [5].

One of the most valuable business applications of Machine Learning is Customer Churn Prediction, which constitutes the primary focus of this report. Organizations in telecommunications, banking, insurance, and subscription-based services use Machine Learning models to identify customers who are likely to discontinue using their products or services. By analyzing customer demographics, service usage, billing history, payment behavior, and customer support interactions, predictive models estimate the probability of churn. Companies can then implement targeted retention campaigns, offer personalized incentives, and improve customer satisfaction before customers decide to leave. Numerous studies have demonstrated that Machine Learning significantly improves customer retention while reducing marketing costs and increasing long-term profitability [6].

The rapid development of Machine Learning continues to create new opportunities across various scientific and industrial domains. As data availability and computational resources continue to increase, Machine Learning is expected to play an even more critical role in supporting intelligent systems, automating complex tasks, and enabling data-driven decision-making in the future [1], [2].

Table 1.5. Major Applications of Machine Learning :

Healthcare	Disease diagnosis, medical image analysis, personalized treatment	Cancer detection, diabetic retinopathy diagnosis
Finance	Fraud detection, credit scoring, financial forecasting	Credit card fraud detection, loan approval
Cybersecurity	Intrusion detection, malware detection, anomaly detection	Network security monitoring
Manufacturing	Predictive maintenance, quality control	Equipment failure prediction
Transportation	Autonomous driving, traffic prediction, route optimization	Self-driving cars, smart logistics
E-commerce	Recommendation systems, customer behavior analysis	Amazon product recommendations, Netflix recommendations
Business Analytics	Customer segmentation, sales forecasting, customer churn prediction	Customer retention strategies

Source: Adapted from Mitchell (1997), Goodfellow et al. (2016), Jordan & Mitchell (2015), and Manzoor et al. (2024).

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https://www.deeplearningbook.org/
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https://doi.org/10.1109/ACCESS.2024.3400045

1.8 Advantages and Limitations of Machine Learning:

Machine Learning has become one of the most powerful technologies in Artificial Intelligence because of its ability to analyze large volumes of data, identify hidden patterns, and make accurate predictions without being explicitly programmed. The widespread adoption of Machine Learning across various industries has demonstrated its effectiveness in solving complex problems and supporting data-driven decision-making. However, despite its remarkable capabilities, Machine Learning also presents several limitations and challenges that must be considered when developing predictive models. Understanding both

the advantages and the limitations of Machine Learning enables organizations to select appropriate algorithms, improve model performance, and ensure reliable decision-making [1], [2].

One of the most significant advantages of Machine Learning is its ability to learn automatically from historical data. Unlike traditional programming approaches that require manually defined rules, Machine Learning algorithms continuously improve their performance as more data become available. This adaptive learning capability enables predictive models to evolve over time and provide increasingly accurate predictions [1].

Another important advantage is the ability of Machine Learning to process massive datasets efficiently. Modern organizations generate enormous amounts of structured and unstructured data every day through customer transactions, social media, sensors, and digital platforms. Machine Learning algorithms can analyze these datasets much faster than manual analytical methods, allowing businesses to extract valuable insights and support strategic decision-making [2].

Machine Learning is also capable of identifying complex relationships and hidden patterns that may not be visible through traditional statistical analysis. Advanced algorithms such as Random Forests, Gradient Boosting, and Artificial Neural Networks can model highly nonlinear relationships between variables, making them particularly suitable for solving complex classification and prediction problems such as Customer Churn Prediction [3].

In addition, Machine Learning contributes significantly to automation and operational efficiency. Intelligent predictive systems reduce human intervention, minimize repetitive tasks, improve decision-making speed, and increase productivity across various sectors, including healthcare, finance, manufacturing, and business analytics [4].

Despite these advantages, Machine Learning also faces several important limitations. One of the primary challenges is its strong dependence on high-quality data. Predictive models learn directly from historical data; therefore, incomplete, inaccurate, or biased datasets may produce unreliable predictions and poor generalization performance [2].

Another limitation concerns computational complexity. Many advanced Machine Learning algorithms, particularly Deep Learning models and Artificial Neural Networks, require powerful computing resources, large memory capacity, and considerable training time. This requirement may increase implementation costs and limit their applicability in organizations with limited computational infrastructure [3].

Machine Learning models may also suffer from overfitting, a situation in which the model performs exceptionally well on training data but poorly on previously unseen observations. Overfitting reduces the model's ability to generalize and often requires techniques such as cross-validation, regularization, and hyperparameter optimization to improve performance [1].

Furthermore, some Machine Learning algorithms, especially Deep Neural Networks, are often considered black-box models because their internal decision-making processes are difficult to interpret. This lack of transparency can create challenges in sectors such as healthcare, banking, and finance, where explainability and accountability are essential for regulatory compliance and ethical decision-making [5].

Consequently, while Machine Learning offers tremendous opportunities for predictive analytics and intelligent decision support, successful implementation requires careful data preparation, appropriate algorithm selection, rigorous model evaluation, and continuous monitoring. Organizations must balance predictive performance with interpretability, computational efficiency, and ethical considerations to maximize the benefits of Machine Learning technologies.

Table 1.6. Advantages and Limitations of Machine Learning:

Learns automatically from data	Requires large amounts of high-quality data

High predictive accuracy	Computationally expensive for complex models
Identifies hidden patterns and relationships	Risk of overfitting
Automates decision-making processes	Difficult to interpret some models (lack Box)
Handles large and complex datasets	Requires continuous model updating
Supports intelligent business decisions	Sensitive to biased or incomplete data

- # CHAPTER 2: ARTIFICIAL NEURAL NETWORKS:

Chapitre 2 : Les Réseaux de Neurones Artificiels (Artificial Neural Networks)

2.1 Introduction to Artificial Neural Networks:

Artificial Neural Networks (ANNs) are among the most powerful and widely used Machine Learning techniques for solving complex prediction and classification problems. Inspired by the structure and functioning of the human brain, Artificial Neural Networks consist of interconnected processing units called artificial neurons, which collaborate to learn patterns and relationships from data. Unlike traditional Machine Learning algorithms that often require manual feature engineering, Neural Networks automatically extract meaningful representations from raw data through multiple layers of computation, enabling them to solve highly nonlinear problems with remarkable accuracy [1].

The development of Artificial Neural Networks originated from the desire to simulate the way biological neurons process and transmit information. The earliest computational neuron model was proposed by McCulloch and Pitts (1943), laying the theoretical foundation for neural computation. Later, the Perceptron, introduced by Frank Rosenblatt (1958), became the first practical learning algorithm capable of performing binary classification. Although early neural network research experienced several periods of stagnation due to computational limitations, the introduction of the Backpropagation algorithm in the 1980s revolutionized the field by enabling efficient training of multi-layer neural networks [2].

In recent years, Artificial Neural Networks have experienced unprecedented growth due to the rapid increase in computational power, the availability of massive datasets, and advances in Graphics Processing Units (GPUs). These developments have led to the emergence of Deep Learning, where neural networks contain multiple hidden layers capable of learning highly complex hierarchical representations of data. Today, Deep Neural Networks have achieved state-of-the-art performance in numerous domains, including computer vision, speech recognition, natural language processing, autonomous driving, financial forecasting, and predictive business analytics [3].

One of the most important business applications of Artificial Neural Networks is Customer Churn Prediction. Organizations in telecommunications, banking, insurance, and subscription-based services use Neural Networks to analyze customer demographics, transaction history, service usage, payment behavior, and customer interactions. By identifying hidden nonlinear relationships among these variables, Neural Networks can accurately estimate the probability that a customer will discontinue a service. These predictions allow companies to implement proactive retention strategies, improve customer satisfaction, and reduce revenue losses associated with customer attrition [4].

Compared with many traditional Machine Learning algorithms, Artificial Neural Networks demonstrate superior performance when dealing with large datasets containing complex interactions among variables. Their ability to automatically learn meaningful features, adapt to new data, and model nonlinear decision boundaries has made them one of the most effective predictive modeling techniques. However, Neural Networks generally require substantial computational resources, careful hyperparameter tuning, and large quantities of labeled training data to achieve optimal performance [3].

The objective of this chapter is to present the theoretical foundations of Artificial Neural Networks, including their biological inspiration, architecture, learning mechanisms, activation functions, major network types, advantages, limitations, and practical applications. Understanding these concepts is essential before implementing the Artificial Neural Network model for Customer Churn Prediction in the practical part of this report.
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[1] S. Haykin, Neural Networks and Learning Machines, 3rd ed. Pearson Education, 2009.
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[3] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016.[3] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016.
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2.2 Biological Inspiration of Artificial Neural Networks:

Artificial Neural Networks (ANNs) are inspired by the structure and functioning of the human brain, particularly the biological neurons that form the nervous system. The human brain contains approximately 86 billion neurons, which communicate with one another through electrical and chemical signals. These neurons work collectively to process information, learn from experience, recognize patterns, make decisions, and control various cognitive and physical functions. The remarkable learning capability of the human brain motivated researchers to develop computational models that mimic its behavior and enable machines to learn from data [1].

A biological neuron consists of several essential components, each performing a specific function during information processing. Dendrites receive signals from neighboring neurons and transmit them to the cell body (soma), where incoming information is integrated. If the accumulated signal exceeds a certain threshold, the neuron generates an electrical impulse that travels along the axon. The impulse is then transmitted to other neurons through specialized junctions known as synapses, where communication occurs via neurotransmitters [2].

Artificial Neural Networks imitate this biological mechanism using mathematical models. In an artificial neuron, input variables correspond to dendrites, while numerical weights represent the strength of synaptic connections. The weighted inputs are summed together, and the result is processed through an activation function to determine the neuron's output. This output is then transmitted to neurons in the next layer of the network. During training, the connection weights are continuously adjusted to minimize prediction errors, allowing the network to learn complex patterns from data [3].

Although Artificial Neural Networks are inspired by biological neurons, they are significantly simplified computational models. Biological neurons communicate through electrochemical processes and exhibit highly complex behaviors that are not fully understood, whereas artificial neurons rely solely on mathematical operations and optimization algorithms. Nevertheless, this simplified architecture has proven highly effective in solving numerous classification, prediction, and pattern recognition problems [4].

The biological inspiration behind Artificial Neural Networks has led to remarkable advances in Artificial Intelligence and Deep Learning. Modern neural networks are capable of

recognizing images, understanding speech, translating languages, driving autonomous vehicles, and predicting customer behavior with exceptional accuracy. Their success demonstrates how concepts borrowed from neuroscience can be transformed into powerful computational techniques capable of solving real-world problems [3].

Table 2.1. Comparison Between Biological and Artificial Neurons

Dendrites receive input signals	Input variables (features
Synapses transmit signal	Weights
Cell body integrates information	Summation function
Axon transmits output signal	Output value
Learning occurs by strengthening synapses	Learning occurs by updating weights

Source: Adapted from Haykin (2009), Goodfellow et al. (2016), and McCulloch & Pitts (1943).

Figure 2.1. Biological Neuron vs Artificial Neuron :

[1] S. Haykin, Neural Networks and Learning Machines, 3rd ed. Pearson Education, 2009.
[1] S. Haykin, Neural Networks and Learning Machines, 3rd ed. Pearson Education, 2009.
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2.3 Architecture of Artificial Neural Networks:

Artificial Neural Networks (ANNs) are composed of interconnected processing units called artificial neurons, which are organized into multiple layers. The architecture of a neural network determines how information flows from the input data to the final output and significantly influences the learning capability and predictive performance of the model. A typical feedforward neural network consists of three main components: the Input Layer, one or more Hidden Layers, and the Output Layer. These layers cooperate to transform raw input data into meaningful predictions through a sequence of mathematical operations [1].

2.3.1 Input Layer

The Input Layer is the first layer of the neural network and serves as the interface between the dataset and the model. Each neuron in this layer represents one input feature from the dataset. Unlike neurons in other layers, input neurons do not perform computations; instead, they simply pass the input values to the next layer.

For Customer Churn Prediction, typical input variables include customer age, gender, contract type, monthly charges, total charges, tenure, internet service, payment method, and customer support history. The number of neurons in the input layer is therefore equal to the number of input features used during training [2].

2.3.2 Hidden Layer

The Hidden Layer is the core computational component of an Artificial Neural Network. It performs complex mathematical transformations that enable the network to discover hidden relationships within the data. Each neuron receives input values from the previous layer, multiplies each input by its corresponding weight, adds a bias term, and applies an

activation function to produce an output.
Modern neural networks may contain multiple hidden layers, giving rise to Deep Neural Networks (DNNs). Increasing the number of hidden layers enables the model to learn more abstract and hierarchical representations of data, thereby improving prediction accuracy for complex problems. However, deeper networks also require more computational resources, larger training datasets, and careful optimization to avoid overfitting [3].
2.3.3 Output Laye:
The Output Layer produces the final prediction generated by the neural network. Its structure depends on the type of learning problem.
For binary classification problems such as Customer Churn Prediction, the output layer usually contains a single neuron with a Sigmoid Activation Function, which produces a probability between 0 and 1. Values close to 1 indicate that the customer is likely to churn, whereas values close to 0 indicate that the customer is expected to remain with the company [2].
For multiclass classification problems, the output layer generally uses the Softmax Activation Function, while regression problems often employ a linear activation function.
2.3.4 Weights and Bias:
Each connection between neurons is associated with a numerical value known as a weight. These weights determine the importance of each input variable during prediction. During training, the neural network continuously updates the weights using optimization algorithms such as Gradient Descent in order to minimize prediction errors.
In addition to weights, each neuron contains a bias term. The bias allows the activation function to shift its output independently of the input values, thereby increasing the flexibility and learning capability of the model [1].
2.3.5 Information Flow:
Information flows sequentially through the network during a process known as Forward Propagation. The input data are transmitted from the Input Layer to the Hidden Layer(s), where mathematical transformations are performed. The processed information then reaches the Output Layer, which generates the final prediction.
During training, prediction errors are propagated backward through the network using the Backpropagation Algorithm. This process updates the weights and biases iteratively until the prediction error is minimized and the model achieves satisfactory performance [4].
Table 2.2. Components of an Artificial Neural Network

Input Layer	Receives the input features from the dataset
Hidden Layer	Learns hidden patterns and nonlinear relationships
Output Layer	Produces the final prediction
Weights	Measure the importance of input connections
Bias	Adjusts the activation threshold of neurons
Activation Function	Introduces nonlinearity into the model

Source: Adapted from Haykin (2009), Goodfellow et al. (2016), and Nielsen (2015).
Figure 2.2. Basic Architecture of an Artificial Neural Network :

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2.4 Types of Artificial Neural Networks:

Artificial Neural Networks (ANNs) can be classified into several types according to their architecture, information flow, and learning mechanism. Each type is designed to solve specific categories of problems and has unique characteristics that make it suitable for particular applications. Among the numerous neural network architectures developed over the years, the most widely used are Feedforward Neural Networks (FFNN), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN). These architectures have become the foundation of modern Artificial Intelligence and Deep Learning applications [1].

2.4.1 Feedforward Neural Network (FFNN)

The Feedforward Neural Network (FFNN) is the simplest and most fundamental type of Artificial Neural Network. In this architecture, information flows in only one direction—from the input layer through one or more hidden layers to the output layer—without any feedback connections or cycles. Each neuron receives input from the previous layer, performs mathematical computations, and forwards the output to the next layer.

Feedforward Neural Networks are mainly used for classification and regression problems. They are particularly suitable for structured datasets where input features are independent. In Customer Churn Prediction, FFNNs analyze customer attributes such as age, contract type, monthly charges, and service usage to estimate the probability that a customer will leave the company. Due to their relatively simple architecture, FFNNs are computationally efficient and serve as the basis for many more advanced neural network models [2].

2.4.2 Convolutional Neural Network (CNN):

Convolutional Neural Networks (CNNs) are specialized neural networks designed primarily for processing image data and other grid-like structures. Unlike Feedforward Neural Networks, CNNs automatically extract spatial features from images through convolutional layers, pooling layers, and fully connected layers.

CNNs have achieved outstanding performance in computer vision tasks such as image classification, facial recognition, object detection, and medical image analysis. They significantly reduce the number of trainable parameters while preserving essential visual information, making them highly efficient for large image datasets. Although CNNs are mainly used in image-related applications, they have also been successfully applied to speech recognition and certain time-series prediction tasks [3].

2.4.3 Recurrent Neural Network (RNN):

Recurrent Neural Networks (RNNs) are specifically designed to process sequential data in which previous information influences future predictions. Unlike Feedforward Neural Networks, RNNs contain feedback connections that allow information from previous time steps to be stored and reused during future computations.

This memory capability makes RNNs particularly effective for sequence modeling tasks such as speech recognition, language translation, handwriting recognition, financial forecasting, and time-series prediction. However, traditional RNNs suffer from the vanishing gradient problem, which limits their ability to learn long-term dependencies. To overcome this limitation, advanced architectures such as Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) were developed [4].

2.4.4 Deep Neural Networks (DNN):

A Deep Neural Network (DNN) is an Artificial Neural Network that contains multiple hidden layers between the input and output layers. Increasing the number of hidden layers enables the network to learn increasingly complex and abstract representations of data. Deep Neural Networks have become the foundation of modern Deep Learning and are

widely applied in natural language processing, speech recognition, autonomous driving, recommendation systems, and predictive analytics. Their ability to model highly nonlinear relationships has made them one of the most accurate approaches for solving complex Machine Learning problems. However, training deep networks requires substantial computational resources, large labeled datasets, and advanced optimization techniques [1], [3].

Comparison of Neural Network Types

Different neural network architectures are designed for different categories of data and applications. Feedforward Neural Networks are suitable for structured tabular data, Convolutional Neural Networks excel at image analysis, Recurrent Neural Networks specialize in sequential data, while Deep Neural Networks provide highly accurate solutions for complex prediction tasks. Selecting the appropriate architecture depends on the characteristics of the dataset, the problem to be solved, and the available computational resources [2].

Table 2.3. Comparison of Artificial Neural Network Types

Advantages	Feedforward Neural Network (FFNN)	Structured data
Classification, Regression, Customer Churn Prediction	Simple architecture, fast training	Convolutional Neural Network (CNN)
Images	Image Classification, Medical Imaging, Object Detection	Excellent feature extraction
Recurrent Neural Network (RNN)	Sequential data	Speech Recognition, Time-Series Forecasting, NLP
Captures temporal dependencies	Deep Neural Network (DNN)	Complex structured and unstructured data
Deep Learning, Prediction, Pattern Recognition	High predictive accuracy	

Source: Adapted from Haykin (2009), Goodfellow et al. (2016), and Nielsen (2015).

[1] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016.[1] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016.
[2] S. Haykin, Neural Networks and Learning Machines, 3rd ed. Pearson Education, 2009.
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[3] M. A. Nielsen, Neural Networks and Deep Learning. Determination Press, 2015[3] M. A. Nielsen, Neural Networks and Deep Learning. Determination Press, 2015
[4] S. Hochreiter and J. Schmidhuber, “Long Short-Term Memory,” Neural Computation, vol. 9, no. 8, pp. 1735–1780, 1997.[4] S. Hochreiter and J. Schmidhuber, “Long Short-Term Memory,” Neural Computation, vol. 9, no. 8, pp. 1735–1780, 1997.

2.5 Learning Process in Artificial Neural Networks:

Artificial Neural Networks learn by continuously adjusting the connection weights between neurons in order to minimize prediction errors. This learning process enables the network to discover hidden relationships within the data and improve its predictive performance over time. The training procedure is generally based on two complementary phases: Forward Propagation and Backpropagation. During these phases, the network repeatedly processes the training data, calculates prediction errors, and updates its parameters until an optimal solution is achieved [1].

2.5.1 Forward Propagation:

Forward Propagation is the first stage of the learning process. In this phase, input data are introduced into the input layer and propagate through the hidden layer(s) until they reach the output layer.

Each neuron performs three main operations:
1. Receives input values from the previous layer.
2. Computes the weighted sum of inputs and bias.
3. Applies an activation function to generate the output.
The mathematical expression of a neuron is:
where:
* x_i = input feature
* w_i = weight
* b = bias
* z = weighted sum
The output of the neuron is then calculated as:
where f represents the activation function.
During Forward Propagation, no learning occurs. The network simply generates predictions using its current weights and biases [2].
2.5.2 Loss Function:
After producing predictions, the network evaluates how close they are to the true target values using a Loss Function (also called the Cost Function).
The objective of training is to minimize this loss.
For binary classification problems such as Customer Churn Prediction, the most commonly used loss function is Binary Cross-Entropy Loss.
A smaller loss value indicates that the model produces more accurate predictions [3].
2.5.3 Backpropagation:
Backpropagation is the learning mechanism that enables Artificial Neural Networks to improve their performance.
After computing the prediction error, the network propagates this error backward from the output layer toward the input layer. Using the chain rule of calculus, the algorithm calculates the gradient of the loss function with respect to every weight and bias.
These gradients indicate how much each parameter contributes to the prediction error.
The weights are then updated according to:
where:
* w_{old} = current weight
* w_{new} = updated weight
* \eta = learning rate
* L = loss function
This process is repeated for every training iteration until the prediction error reaches an acceptable level [4].
2.5.4 Gradient Descent:
Gradient Descent is the optimization algorithm responsible for updating the weights during Backpropagation.
Its objective is to minimize the loss function by moving the weights in the direction that reduces prediction error.
Several variants of Gradient Descent exist:
* Batch Gradient Descent
* Stochastic Gradient Descent (SGD)
* Mini-Batch Gradient Descent
Among these, Mini-Batch Gradient Descent is the most commonly used in Deep Learning because it provides an excellent balance between computational efficiency and convergence speed [3].
2.5.5 Learning Rate:
The Learning Rate is one of the most important hyperparameters in Artificial Neural Networks.

It determines the size of each weight update during training.
* A very small learning rate results in slow convergence.
* A very large learning rate may prevent convergence or cause unstable learning.
Selecting an appropriate learning rate is therefore essential for obtaining high predictive performance [2].
Training Cycle
The complete learning process is repeated for multiple epochs until the model converges. Each epoch consists of:
1. Forward Propagation
2. Prediction
3. Loss Calculation
4. Backpropagation
5. Weight Update
This iterative optimization process gradually improves the prediction accuracy of the Artificial Neural Network.
Table 2.4. Learning Process in Artificial Neural Networks

Forward Propagation	Computes predictions
Loss Function	Measures prediction error
Backpropagation	Calculates gradients
Gradient Descent	Updates weights
Repeat (Epochs)	Improves prediction accuracy

Figure 2.3 – Illustration du processus d'apprentissage par rétropropagation
Source: Adapted from Goodfellow et al. (2016), Haykin (2009), and Nielsen (2015).
[1] S. Haykin, Neural Networks and Learning Machines, 3rd ed. Pearson Education, 2009.
[1] S. Haykin, Neural Networks and Learning Machines, 3rd ed. Pearson Education, 2009.
[2] M. A. Nielsen, Neural Networks and Deep Learning. Determination Press, 2015.[2] M. A. Nielsen, Neural Networks and Deep Learning. Determination Press, 2015.
[3] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016.[3] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016.
[4] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, “Learning Representations by Back-Propagating Errors,” Nature, vol. 323, no. 6088, pp. 533–536, 1986.[4] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, “Learning Representations by Back-Propagating Errors,” Nature, vol. 323, no. 6088, pp. 533–536, 1986.
2.6 Activation Functions:
Activation Functions are mathematical functions applied to the output of each neuron in an Artificial Neural Network. Their primary purpose is to introduce nonlinearity into the network, allowing it to learn complex relationships between input variables and target outputs. Without activation functions, a neural network would behave like a simple linear model regardless of the number of hidden layers, making it incapable of solving complex classification and prediction problems [1].
An activation function transforms the weighted sum of inputs into an output value that is passed to the next layer. The selection of an appropriate activation function has a significant impact on the convergence speed, learning efficiency, and predictive performance of the neural network. Different activation functions are suitable for different tasks depending on the architecture of the network and the type of learning problem [2].
2.6.1 Sigmoid Function:
The Sigmoid activation function is one of the earliest activation functions used in Artificial

Neural Networks. It transforms any input value into a probability between 0 and 1, making it particularly suitable for binary classification problems.
The mathematical expression of the Sigmoid function is:
The output approaches 0 for very negative values and 1 for very positive values.
Advantages
* Produces probability values.
* Easy to interpret.
* Suitable for binary classification.
Limitations
* Suffers from the vanishing gradient problem.
* Slow convergence in deep neural networks.
In Customer Churn Prediction, the output layer commonly uses the Sigmoid function because the prediction represents the probability that a customer will churn.
2.6.2 Hyperbolic Tangent (Tanh):
The Tanh activation function is similar to Sigmoid but produces outputs between −1 and +1.
Its mathematical expression is:
Unlike Sigmoid, Tanh is zero-centered, which often leads to faster convergence during training.
Advantages
* Zero-centered outputs.
* Faster convergence than Sigmoid.
Limitations
* Still affected by the vanishing gradient problem.
2.6.3 ReLU (Rectified Linear Unit):
The Rectified Linear Unit (ReLU) is currently the most widely used activation function in Deep Learning.
Its mathematical expression is:
If the input is negative, the output is 0.
If the input is positive, the output equals the input value.
Advantages:
* Computationally efficient.
* Reduces the vanishing gradient problem.
* Accelerates model training.
Limitations
* May suffer from the Dead ReLU Problem, where some neurons stop learning permanently.
For this reason, ReLU is commonly used in the hidden layers of Artificial Neural Networks.
2.6.4 Leaky ReLU:
Leaky ReLU is an improved version of ReLU.
Instead of assigning zero to all negative inputs, Leaky ReLU allows a small negative slope.
Advantages
* Solves the Dead ReLU problem.
* Maintains fast convergence.
Limitations
* Requires selecting the leakage coefficient.
2.6.5 Softmax Function:
The Softmax activation function is mainly used in multiclass classification.
It converts output values into probabilities whose sum equals 1.
The class with the highest probability becomes the final prediction.
Softmax is not generally used for Customer Churn Prediction because churn prediction is a binary classification problem.

Table 2.5. Comparison of Activation Functions

Sigmoid	0 to 1	Probability output	Binary Classification
Tanh	-1 to +1	Zero-centered	Hidden Layers
ReLU	0 to +∞	Fast training	Deep Neural Networks
Leaky ReLU	-∞ to +∞	Prevents dead neurons	Deep Learning
Softmax	0 to 1	Multi-class probabilities	Multi-class Classification

Figure 2.4 – Illustration des fonctions d'activation usuelles
Source: Adapted from Goodfellow et al. (2016), Haykin (2009), and Nielsen (2015).
Which Activation Functions Are Used in This Project?
For the Customer Churn Prediction model developed in this report:
* Hidden Layers: ReLU Activation Function.
* Output Layer: Sigmoid Activation Function.
This combination provides high predictive accuracy and efficient training for binary classification tasks
[1] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016.[1] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016.
[2] S. Haykin, Neural Networks and Learning Machines, 3rd ed. Pearson Education, 2009.
[2] S. Haykin, Neural Networks and Learning Machines, 3rd ed. Pearson Education, 2009.
[3] M. A. Nielsen, Neural Networks and Deep Learning. Determination Press, 2015.[3] M. A. Nielsen, Neural Networks and Deep Learning. Determination Press, 2015.
2.7 Advantages and Limitations of Artificial Neural Networks:
Artificial Neural Networks (ANNs) have become one of the most powerful techniques in Artificial Intelligence and Machine Learning due to their remarkable ability to learn complex relationships from data. Their flexibility, adaptability, and predictive capabilities have enabled their widespread adoption across numerous scientific and industrial domains. Despite these strengths, Artificial Neural Networks also present several limitations that should be considered when designing and implementing predictive models. Understanding both the advantages and the limitations of ANNs is essential for selecting appropriate network architectures and achieving reliable model performance [1].
One of the most significant advantages of Artificial Neural Networks is their ability to model highly nonlinear relationships between variables. Unlike traditional statistical methods that often assume linear relationships, neural networks automatically learn complex patterns directly from the training data. This capability makes them particularly effective for solving difficult prediction and classification problems such as Customer Churn Prediction, fraud detection, and medical diagnosis [2].
Another important advantage is the automatic feature learning capability of ANNs. Traditional Machine Learning algorithms often require manual feature engineering before training. In contrast, Artificial Neural Networks automatically extract relevant features from raw data during the learning process, reducing human intervention and improving predictive performance [3].
Artificial Neural Networks are also highly adaptive. As new training data become available, the network can continue learning by updating its internal weights and biases. This adaptive behavior enables ANNs to improve continuously and remain effective in dynamic environments where customer behavior and business conditions frequently change [1].
Furthermore, Artificial Neural Networks provide high predictive accuracy, especially when trained on large datasets. Advances in optimization algorithms, activation functions, and

computational hardware have significantly improved their performance in applications involving image recognition, speech processing, financial forecasting, and business analytics [3].

Despite these advantages, Artificial Neural Networks also have several limitations. One of the primary challenges is their requirement for large amounts of high-quality training data. Insufficient or noisy datasets may reduce model accuracy and increase the risk of poor generalization [2].

Another limitation concerns the computational cost associated with training neural networks. Large models containing many hidden layers require substantial memory, powerful processors, and long training times. This can increase implementation costs, particularly for organizations with limited computational resources [3].

Artificial Neural Networks are often criticized for their lack of interpretability. Because the internal decision-making process involves numerous hidden neurons and mathematical transformations, it is often difficult to explain how specific predictions are generated. Consequently, ANNs are frequently referred to as black-box models, which may limit their adoption in highly regulated industries such as healthcare and finance, where explainability is essential [4].

Another challenge is the risk of overfitting. When a neural network becomes excessively complex or is trained for too many iterations, it may memorize the training data instead of learning general patterns. As a result, the model performs well on the training dataset but poorly on previously unseen data. Various regularization techniques, dropout layers, and early stopping methods are commonly used to reduce overfitting and improve generalization [3].

Overall, Artificial Neural Networks offer exceptional predictive capabilities and have become indispensable tools for solving complex real-world problems. However, their successful implementation requires careful data preparation, appropriate network design, hyperparameter tuning, and continuous performance evaluation. Balancing predictive accuracy with computational efficiency and model interpretability remains one of the most important challenges in modern Artificial Intelligence research.

Table 2.6. Advantages and Limitations of Artificial Neural Networks:

Source: Adapted from Haykin (2009), Goodfellow et al. (2016), and Nielsen (2015).

[1] S. Haykin, Neural Networks and Learning Machines, 3rd ed. Pearson Education, 2009.
[1] S. Haykin, Neural Networks and Learning Machines, 3rd ed. Pearson Education, 2009.
[2] C. M. Bishop, Pattern Recognition and Machine Learning. Springer, 2006.[2] C. M. Bishop, Pattern Recognition and Machine Learning. Springer, 2006.
[3] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016.[3] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016.
[4] M. A. Nielsen, Neural Networks and Deep Learning. Determination Press, 2015.[4] M. A. Nielsen, Neural Networks and Deep Learning. Determination Press, 2015.

2.8 Applications of Artificial Neural Networks:

Artificial Neural Networks (ANNs) have become one of the most influential technologies in Artificial Intelligence due to their ability to learn complex patterns from large datasets and generate highly accurate predictions. Their flexibility and adaptability have enabled their successful application across numerous scientific, industrial, and commercial domains. Today, Artificial Neural Networks are widely employed in healthcare, finance, manufacturing, transportation, cybersecurity, natural language processing, and business analytics. As computational resources and data availability continue to increase, ANN-based solutions are becoming increasingly important in solving complex real-world problems [1].

One of the most significant application areas of Artificial Neural Networks is healthcare. Neural Networks assist medical professionals in diagnosing diseases, analyzing medical

images, predicting disease progression, and recommending personalized treatment plans. Deep Neural Networks, particularly Convolutional Neural Networks (CNNs), have demonstrated exceptional performance in detecting breast cancer, diabetic retinopathy, pneumonia, and other diseases from medical images. These systems improve diagnostic accuracy while reducing the workload of healthcare professionals [2].

In the financial sector, Artificial Neural Networks are extensively used for fraud detection, credit risk assessment, stock market forecasting, and financial decision support. Banks analyze millions of daily financial transactions using ANN models to detect suspicious activities and identify fraudulent behavior in real time. Additionally, Neural Networks support credit scoring by evaluating customer financial profiles and predicting the likelihood of loan repayment [3].

Artificial Neural Networks also play a crucial role in manufacturing and industrial automation. Modern industries utilize ANN-based predictive maintenance systems to monitor industrial equipment, analyze sensor data, and predict machine failures before they occur. This proactive maintenance strategy minimizes production downtime, reduces maintenance costs, and improves operational efficiency [1].

Another important application area is transportation and autonomous vehicles. Self-driving vehicles rely on Artificial Neural Networks to recognize traffic signs, detect pedestrians, identify obstacles, and make real-time driving decisions. Neural Networks process information collected from cameras, radar, and LiDAR sensors, enabling autonomous navigation under dynamic road conditions [2].

Artificial Neural Networks have significantly transformed the field of Natural Language Processing (NLP). They are used in machine translation, sentiment analysis, speech recognition, question-answering systems, and intelligent virtual assistants. Advanced neural architectures such as Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and Transformer-based models have dramatically improved the performance of language understanding systems [3].

One of the most valuable business applications of Artificial Neural Networks is Customer Churn Prediction, which constitutes the primary focus of this report. Companies operating in telecommunications, banking, insurance, and subscription-based services continuously analyze customer behavior using ANN models to estimate the probability that customers will discontinue their services. Variables such as customer demographics, service usage, payment history, contract type, customer complaints, and interaction history are processed simultaneously to identify customers at high risk of churn. Based on these predictions, organizations implement targeted retention campaigns, personalized offers, and proactive customer support strategies to reduce churn and improve customer loyalty. Numerous studies have demonstrated that Artificial Neural Networks outperform many traditional Machine Learning algorithms in customer churn prediction due to their ability to model highly nonlinear relationships among customer attributes [4].

Artificial Neural Networks are also widely used in recommendation systems. E-commerce platforms, streaming services, and online retailers analyze customer preferences and purchasing behavior to recommend products, movies, music, and personalized content. Companies such as Amazon, Netflix, and Spotify employ advanced neural network models to improve customer satisfaction and enhance user experience [2].

The continuous advancement of Artificial Intelligence, cloud computing, and high-performance computing technologies is expected to further expand the application domains of Artificial Neural Networks. Their ability to process large-scale data, automate decision-making, and generate highly accurate predictions ensures that ANNs will remain one of the most important technologies driving digital transformation across industries in the coming years.

Table 2.7. Major Applications of Artificial Neural Networks:

Healthcare	Disease diagnosis, medical image analysis	Cancer detection, MRI analysis
Finance	Fraud detection, credit scoring, stock prediction	Fraud detection systems
Manufacturing	Predictive maintenance, quality inspection	Equipment failure prediction
Transportation	Autonomous driving, traffic prediction	Self-driving vehicles
Natural Language Processing	Machine translation, speech recognition	Chatbots, virtual assistants
Business Analytics	Customer churn prediction, sales forecasting	Customer retention strategies
Recommendation Systems	Personalized recommendations	Netflix, Amazon, Spotify

Source: Adapted from Goodfellow et al. (2016), Haykin (2009), Bishop (2006), and Manzoor et al. (2024).

Conclusion du chapitre 2:

Artificial Neural Networks represent one of the most powerful Machine Learning techniques for solving complex prediction and classification problems. Inspired by the biological structure of the human brain, ANNs are capable of automatically learning hidden patterns and nonlinear relationships from large datasets. This chapter presented the theoretical foundations of Artificial Neural Networks, including their biological inspiration, architecture, major network types, learning mechanisms, activation functions, advantages, limitations, and practical applications. Among the various application domains, Customer Churn Prediction stands out as a critical business problem where Artificial Neural Networks provide highly accurate predictive models that support customer retention and strategic decision-making. The theoretical concepts discussed in this chapter establish the foundation for the practical implementation presented in the next chapter, where an Artificial Neural Network model will be developed and evaluated using Google Colab for predicting customer churn.

[1] S. Haykin, Neural Networks and Learning Machines, 3rd ed. Pearson Education, 2009.
[1] S. Haykin, Neural Networks and Learning Machines, 3rd ed. Pearson Education, 2009.
[2] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016.[2] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016.
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Prediction and Recommendations for Business Practitioners,” IEEE Access, vol. 12, pp. 70434–70463, 2024.

PARTIE II — MISE EN ŒUVRE PRATIQUE

Chapitre 3 : Prédiction du Churn Client à l'aide d'un Réseau de Neurones — Implémentation Pratique:

Cette seconde partie prolonge le cadre théorique présenté au chapitre précédent en mettant en œuvre, de manière concrète, un modèle de réseau de neurones artificiel pour la prédiction du churn client. Le notebook analysé (Untitled6 (1).ipynb) applique directement les

concepts exposés dans la partie théorique : prétraitement des données tabulaires, architecture Dense/Dropout/Sigmoid, fonction de perte binary_crossentropy, optimiseur Adam, et évaluation par matrice de confusion, rapport de classification et courbe ROC-AUC.

3.1 Présentation générale du notebook:

Domaine : Machine Learning – Deep Learning. Problème : prédiction du churn client. Jeu de données : Telco Customer Churn (7 043 observations initiales). Méthode : prétraitement tabulaire suivi d'un classifieur binaire par réseau de neurones. Sortie du modèle : probabilité de churn, classification binaire et niveau de risque. Le notebook implémente un pipeline complet d'apprentissage supervisé : chargement des données brutes, nettoyage, prétraitement, construction du réseau de neurones, entraînement, évaluation sur données non vues, visualisation des résultats, test d'une prédiction individuelle et sauvegarde du modèle entraîné.

Table 3.1 – Synthèse pratique du pipeline et des résultats du notebook

Dimensions initiales	7043 lignes × 21 colonnes
Variable cible	Churn : Yes/No, encodée Yes → 1 et No → 0
Clients non-churners	5174 clients
Clients churners	1869 clients
Taux de churn initial	26,54 %
Lignes après nettoyage	7032 lignes
Variables explicatives avant encodage	19 variables
Variables d'entrée après prétraitement	45 variables numériques
Précision sur le jeu de test (Test accuracy)	78,46 %
ROC-AUC	environ 0,82

Figure 3.1 – Distribution de la variable cible Churn

Observation clé : la variable cible est déséquilibrée : 5174 clients n'ont pas résilié, contre 1869 clients churners. En raison de ce déséquilibre, l'accuracy seule n'est pas suffisante ; le rappel de la classe churn et la matrice de confusion doivent également être analysés.

3.2 Analyse détaillée de toutes les étapes pratiques:

Cette section suit l'ordre exact du notebook. Chaque étape est décrite à travers son objectif, les opérations effectuées, les résultats observés et une interprétation pratique de son rôle dans le pipeline de Machine Learning.

3.2.1 Importation des librairies:

Objectif : préparer l'environnement Python et charger les outils nécessaires à la manipulation des données, au prétraitement, à la modélisation, à la visualisation et à l'évaluation.

Opérations effectuées : Le notebook importe pandas et numpy pour la manipulation des données, matplotlib pour la visualisation, scikit-learn pour le découpage, le prétraitement et l'évaluation, ainsi que TensorFlow/Keras pour la modélisation du réseau de neurones.

Résultats observés : La sortie du notebook confirme la version 2.20.0 de TensorFlow, ce qui garantit la disponibilité de l'environnement de Deep Learning.

Interprétation pratique : Cette étape établit la pile technique utilisée tout au long de l'implémentation. Keras est adapté à la construction d'un réseau de neurones dense simple pour une classification binaire.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
import tensorflow as tf
from tensorflow import keras
```

3.2.2 Chargement du jeu de données:

Objectif : charger le jeu de données Telco Customer Churn dans un DataFrame pandas.
Opérations effectuées : Le jeu de données est chargé depuis une URL GitHub à l'aide de pd.read_csv, puis df.head() est utilisé pour afficher les premières lignes.
Résultats observés : Le DataFrame initial contient 7043 observations et 21 colonnes. Les premières lignes montrent des types de variables mixtes : numériques, textuelles binaires et catégorielles.
Interprétation pratique : Le chargement direct depuis une URL rend le notebook reproductible tant que la source reste disponible. L'affichage des premières lignes permet une vérification initiale de la structure des données.

```
url = "https://raw.githubusercontent.com/IBM/telco-customer-churn-on-icp4d/master/data/Telco-Customer-Churn.csv"
df = pd.read_csv(url)
df.head()
```

3.2.3 Exploration des données:

Objectif : comprendre la structure du jeu de données, les colonnes disponibles, les types de données, les statistiques descriptives et la distribution de la cible.
Opérations effectuées : Le notebook utilise df.shape, df.columns, df.info(), df.describe(), value_counts() sur Churn, et un diagramme en barres pour la variable cible.
Résultats observés : La cible contient 5174 valeurs No et 1869 valeurs Yes. Les statistiques descriptives montrent initialement des colonnes numériques telles que SeniorCitizen, tenure et MonthlyCharges avant la conversion de TotalCharges.
Interprétation pratique : Cette étape révèle le déséquilibre des classes et souligne la nécessité de traiter différemment les variables numériques et catégorielles.

```
df.shape
df.columns
df.info()
df.describe()
df["Churn"].value_counts()
```

3.2.4 Nettoyage des données:

Objectif : corriger les types de données incohérents, supprimer les enregistrements problématiques et éliminer les variables non prédictives.
Opérations effectuées : TotalCharges est converti en numérique via pd.to_numeric avec errors="coerce". Les lignes avec des valeurs manquantes de TotalCharges sont supprimées. customerID est supprimé. La cible Churn est convertie de Yes/No vers 1/0.
Résultats observés : Après nettoyage, le jeu de données contient 7032 observations. La variable cible devient numérique et peut être utilisée directement pour la classification binaire.
Interprétation pratique : La conversion de TotalCharges est nécessaire car les réseaux de neurones requièrent des entrées numériques. La suppression de customerID évite d'introduire un identifiant sans signification prédictive généralisable.

```
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
df.dropna(inplace=True)
df.drop("customerID", axis=1, inplace=True)
df["Churn"] = df["Churn"].map({"Yes": 1, "No": 0})
```

3.2.5 Séparation des variables explicatives et de la cible:

Objectif : créer la matrice explicative X et le vecteur cible y.

Opérations effectuées : Le notebook retire Churn de l'ensemble des variables et le stocke séparément sous y.
Résultats observés : X contient 19 variables explicatives, tandis que y contient la cible binaire encodée.
Interprétation pratique : Cette séparation est nécessaire avant l'entraînement, car le modèle doit apprendre la relation entre les variables d'entrée et la cible plutôt que de recevoir la cible comme partie de l'entrée.

```
X = df.drop("Churn", axis=1)
y = df["Churn"]
```

3.2.6 Identification des colonnes numériques et catégorielles:
Objectif : séparer les variables numériques des variables catégorielles afin que chaque groupe reçoive la transformation de prétraitement adéquate.
Opérations effectuées : Le notebook utilise select_dtypes pour extraire les colonnes numériques et les colonnes catégorielles de type objet.
Résultats observés : Les variables numériques sont SeniorCitizen, tenure, MonthlyCharges et TotalCharges. Les autres variables explicatives textuelles sont traitées comme catégorielles.
Interprétation pratique : Cette distinction est essentielle car les réseaux de neurones opèrent sur des matrices numériques. Les colonnes catégorielles doivent être encodées, tandis que les colonnes numériques doivent être standardisées.

```
numeric_cols = X.select_dtypes(include=["int64", "float64"]).columns
categorical_cols = X.select_dtypes(include=["object"]).columns
```

3.2.7 Découpage en ensembles d'entraînement et de test:
Objectif : créer un ensemble d'entraînement pour l'apprentissage des paramètres du modèle et un ensemble de test distinct pour évaluer la généralisation.
Opérations effectuées : Le notebook applique train_test_split avec test_size=0.2, random_state=42 et stratify=y.
Résultats observés : Le découpage produit X_train de forme (5625, 19), X_test de forme (1407, 19), y_train de forme (5625,) et y_test de forme (1407,).
Interprétation pratique : L'utilisation de stratify=y est importante car elle préserve la proportion churn/non-churn dans les deux sous-ensembles, réduisant le biais d'évaluation causé par le déséquilibre des classes.

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

3.2.8 Prétraitement des données:
Objectif : transformer le jeu de données brut en une matrice entièrement numérique utilisable par le réseau de neurones.
Opérations effectuées : Un ColumnTransformer applique StandardScaler aux variables numériques et OneHotEncoder(handle_unknown="ignore") aux variables catégorielles. Le préprocesseur est ajusté sur X_train puis appliqué à X_train et X_test.
Résultats observés : Après prétraitement, la matrice d'entrée contient 45 variables numériques.
Interprétation pratique : La standardisation évite que les variables à plus grande échelle numérique dominent le processus d'optimisation. L'encodage one-hot convertit les valeurs catégorielles en indicateurs binaires adaptés à Keras.

```
preprocessor = ColumnTransformer(
    transformers=[
        ("num", StandardScaler(), numeric_cols),
        ("cat", OneHotEncoder(handle_unknown="ignore"), categorical_cols)
    ]
```

```
)
X_train_processed = preprocessor.fit_transform(X_train)
X_test_processed = preprocessor.transform(X_test)
```

3.2.9 Construction de l'architecture du réseau de neurons:
Objectif : définir le modèle de classification binaire.
Opérations effectuées : Le notebook crée un modèle Sequential avec une couche d'entrée de 45 variables, deux couches cachées Dense de 64 et 32 neurones, deux couches Dropout, et un neurone de sortie sigmoïde.
Résultats observés : Selon le résumé Keras, le modèle contient 5 057 paramètres entraînables.
Interprétation pratique : Les activations ReLU permettent au réseau d'apprendre des relations non linéaires. La sortie sigmoïde produit une probabilité de churn. Le Dropout aide à réduire le surapprentissage.

```
model = keras.Sequential([
    layers.Input(shape=(X_train_processed.shape[1],)),
    layers.Dense(64, activation="relu"),
    layers.Dropout(0.3),
    layers.Dense(32, activation="relu"),
    layers.Dropout(0.2),
    layers.Dense(1, activation="sigmoid")
])
```

3.2.10 Compilation du modèle:
Objectif : définir la stratégie d'optimisation, la fonction de perte et la métrique surveillée.
Opérations effectuées : Le modèle est compilé avec optimizer="adam", loss="binary_crossentropy", et metrics=["accuracy"].
Résultats observés : Après compilation, le modèle est prêt pour l'entraînement.
Interprétation pratique : Adam est un optimiseur robuste pour l'entraînement basé sur le gradient. La binary crossentropy est adaptée à une sortie sigmoïde et une cible binaire.

```
model.compile(
    optimizer="adam",
    loss="binary_crossentropy",
    metrics=["accuracy"]
)
```

3.2.11 Entraînement du modèle:
Objectif : ajuster les poids du réseau de neurones à l'aide des données d'entraînement.
Opérations effectuées : Le notebook entraîne le modèle sur 50 époques, avec batch_size=32 et validation_split=0.2.
Résultats observés : La meilleure accuracy de validation est de 81,07 % à l'époque 5, avec une perte de validation de 0,4003. À l'époque 50, l'accuracy d'entraînement est de 82,44 % et l'accuracy de validation de 78,49 %.
Interprétation pratique : Les journaux montrent que la performance de validation atteint son meilleur niveau relativement tôt, puis se stabilise ou diminue légèrement. Cela suggère qu'un EarlyStopping pourrait améliorer le processus d'entraînement.

```
history = model.fit(
    X_train_processed, y_train,
    validation_split=0.2,
    epochs=50,
    batch_size=32,
    verbose=1
)
```

3.2.12 Évaluation du modèle sur l'ensemble de test:
Objectif : mesurer la performance sur des données non utilisées durant l'entraînement.
Opérations effectuées : Le notebook utilise model.evaluate sur X_test_processed et y_test

pour calculer la perte et l'accuracy.
Résultats observés : La perte de test est de 0,4523 et l'accuracy de test de 78,46 %.
Interprétation pratique : Cette performance globale est acceptable comme référence initiale, mais l'accuracy doit être interprétée avec prudence car les classes cibles sont déséquilibrées.
loss, accuracy = model.evaluate(X_test_processed, y_test)

3.2.13 Génération des prédictions et application d'un seuil de classification:
Objectif : convertir les probabilités prédites en classes binaires.
Opérations effectuées : Le notebook prédit les probabilités avec model.predict et applique un seuil fixe de 0,5. Les probabilités supérieures à 0,5 deviennent la classe 1 ; sinon la classe 0.
Résultats observés : L'accuracy recalculée avec accuracy_score est de 78,46 %, cohérente avec le résultat de l'évaluation du modèle.
Interprétation pratique : Le seuil de 0,5 est standard, mais il n'est peut-être pas optimal pour un objectif de rétention. Abaisser le seuil peut détecter davantage de churners, au prix d'une augmentation des faux positifs.
y_pred_prob = model.predict(X_test_processed)
y_pred = (y_pred_prob > 0.5).astype(int)

3.2.14 Construction de la matrice de confusion:
Objectif : décomposer les prédictions correctes et incorrectes en vrais négatifs, faux positifs, faux négatifs et vrais positifs.
Opérations effectuées : le notebook calcule confusion_matrix(y_test, y_pred).
Résultats observés : la matrice est [[895, 138], [165, 209]], correspondant à 895 vrais négatifs, 138 faux positifs, 165 faux négatifs et 209 vrais positifs.
Interprétation pratique : les 165 faux négatifs sont critiques dans la prédiction du churn car ce sont des churners réels classés à tort comme non-churners.
Table 3.2 – Matrice de confusion numérique

Actual No Churn	895	138
Actual Churn	165	209

Figure 3.2 – Visualisation de la matrice de confusion
3.2.15 Production du rapport de classification:
Objectif : évaluer la précision, le rappel et le F1-score séparément pour chaque classe.
Opérations effectuées : le notebook utilise classification_report(y_test, y_pred).
Résultats observés : pour la classe 0, précision = 0,84, rappel = 0,87, F1-score = 0,86. Pour la classe 1, précision = 0,60, rappel = 0,56, F1-score = 0,58.
Interprétation pratique : le modèle reconnaît mieux la classe majoritaire No Churn. Le rappel churn reste modéré, ce qui limite le modèle pour une rétention proactive.
Table 3.3 – Rapport de classification issu du notebook

0 – No Churn	0.84	0.87	0.86	1033
1 – Churn	0.60	0.56	0.58	374
Accuracy	-	-	0.78	1407
Macro avg	0.72	0.71	0.72	1407
Weighted avg	0.78	0.78	0.78	1407

3.2.16 Visualisation de l'accuracy et de la perte:
Objectif : analyser la dynamique d'apprentissage et la relation entre performance

d'entraînement et de validation.
Opérations effectuées : le notebook trace l'accuracy d'entraînement, l'accuracy de validation, la perte d'entraînement et la perte de validation.
Résultats observés : l'accuracy de validation atteint son maximum à l'époque 5 (81,07 %), puis diminue légèrement jusqu'à l'époque 50.
Interprétation pratique : l'écart croissant entre les comportements d'entraînement et de validation indique un possible surapprentissage. Un EarlyStopping serait une amélioration pertinente.

Figure 3.3 – Accuracy d'entraînement et de validation

Figure 3.4 – Perte d'entraînement et de validation
3.2.17 Test d'un exemple client individuel:
Objectif : illustrer l'utilisation du modèle entraîné sur une observation client unique.
Opérations effectuées : le notebook sélectionne la première observation de X_test, applique le préprocesseur ajusté, prédit la probabilité de churn et affiche une décision textuelle.
Résultats observés : la probabilité de churn prédite est de 0,0051, soit 0,51 %. Étant inférieure à 0,5, le client est classé comme peu susceptible de résilier.
Interprétation pratique : cette étape transforme le modèle en outil de scoring individuel. La probabilité prédite peut appuyer une décision pratique de rétention.
3.2.18 Analyse ROC:
Objectif : évaluer la capacité du modèle à distinguer les churners des non-churners selon différents seuils de décision.
Opérations effectuées : le notebook calcule roc_curve et auc à partir des probabilités prédites, puis trace la courbe ROC.
Résultats observés : l'AUC affichée est d'environ 0,82.
Interprétation pratique : une AUC d'environ 0,82 indique une capacité discriminative satisfaisante. Cependant, l'AUC ne détermine pas le seuil opérationnel ; celui-ci doit être choisi selon le coût métier des faux positifs et des faux négatifs.

Figure 3.5 – Courbe ROC du modèle
3.2.19 Sauvegarde du modèle entraîné:
Objectif : conserver le réseau de neurones entraîné pour une réutilisation future.
Opérations effectuées : le notebook sauvegarde le modèle avec model.save("churn_model.h5").
Résultats observés : Keras affiche un avertissement indiquant que le format HDF5 .h5 est considéré comme obsolète et recommande le format natif .keras.
Interprétation pratique : la sauvegarde du modèle est nécessaire à sa réutilisation ou à son déploiement. Pour un projet Keras récent, model.save("churn_model.keras") serait préférable.
3.2.20 Attribution d'un niveau de risque:
Objectif : convertir la probabilité de churn prédite en une catégorie de risque métier lisible.
Opérations effectuées : le notebook définit trois niveaux de risque : High Risk si probabilité > 0,7, Medium Risk si probabilité > 0,4, et Low Risk sinon.
Résultats observés : pour l'exemple testé, le résultat est Low Risk car la probabilité prédite n'est que de 0,51 %.
Interprétation pratique : cette règle rend la sortie du modèle plus compréhensible pour un utilisateur métier. Cependant, les seuils 0,7 et 0,4 sont arbitraires et devraient être calibrés selon les objectifs de l'entreprise.
3.3 Analyse pratique des résultats:
Le réseau de neurones atteint une accuracy de test de 78,46 %. Cela signifie que le modèle

classe correctement environ 78 clients sur 100 dans l'ensemble de test. Cependant, la distribution de la cible étant déséquilibrée, cette métrique globale doit être analysée conjointement avec les métriques par classe.

Table 3.4 – Interprétation pratique des principales métriques d'évaluation

Test accuracy	78,46 %	Performance globale acceptable comme référence.
Test loss	0,4523	Erreur de prédiction moyenne sur l'ensemble de test.
Précision churn	60,23 %	Environ 60 % des clients prédits comme churners le sont réellement.
Rappel churn	55,88 %	Le modèle détecte un peu plus de la moitié des churners réels.
F1-score churn	57,98 %	Compromis modéré pour la classe positive.
Spécificité No Churn	86,64 %	Le modèle identifie relativement bien les non-churners.
Taux de faux négatifs	44,12 %	Une part importante des churners est manquée.
ROC-AUC	0,82	La capacité de discrimination globale est satisfaisante.

Limitation pratique la plus importante : le modèle manque 165 churners réels dans l'ensemble de test. Dans un contexte réel de rétention client, ces faux négatifs sont coûteux car l'entreprise ne ciblerait pas ces clients avec une action de rétention.

3.4 Forces et limites pratiques:

3.4.1 Forces du pipeline implémenté:

Le notebook suit un pipeline de Machine Learning supervisé complet et cohérent.

Le prétraitement est adapté aux données tabulaires mixtes : standardisation pour les variables numériques et encodage one-hot pour les variables catégorielles.

Le découpage train/test est stratifié, ce qui est pertinent pour une cible déséquilibrée.

Le réseau de neurones utilise des couches Dropout pour réduire le risque de surapprentissage.

L'évaluation inclut l'accuracy, la matrice de confusion, le rapport de classification, les courbes d'apprentissage et la ROC-AUC.

Le notebook inclut un exemple de prédiction individuelle, la sauvegarde du modèle et des catégories de risque orientées métier.

3.4.2 Limites du pipeline implémenté:

La classe Churn est minoritaire : 1869 churners contre 5174 non-churners dans le jeu de données initial.

Le rappel churn n'est que de 55,88 %, ce qui signifie que 165 churners sont manqués dans l'ensemble de test.

Le seuil de classification de 0,5 n'est pas optimisé pour un objectif de rétention.

Les résultats de validation suggèrent qu'un entraînement de 50 époques sans EarlyStopping peut conduire à un surapprentissage.

Le notebook n'inclut pas d'analyse d'importance des variables ou d'explicabilité du modèle.

Le format de sauvegarde .h5 est signalé par Keras comme obsolète.

3.5 Recommandations pratiques d'amélioration:

Table 3.5 – Recommandations pratiques d'amélioration

Surapprentissage	Ajouter EarlyStopping avec restore_best_weights=True.	Conserver le meilleur modèle de validation.
Déséquilibre des classes	Tester class_weight, sur-échantillonnage ou sous-échantillonnage.	Améliorer le rappel churn.
Seuil de décision	Sélectionner le seuil selon le F1, le rappel ou le coût métier.	Réduire les faux négatifs lorsque la rétention est prioritaire.
Métriques	Ajouter la courbe précision-rappel et la courbe de calibration.	Évaluer plus précisément la qualité des probabilités.
Validation	Utiliser la validation croisée ou des découpages répétés.	Obtenir une estimation de performance plus robuste.
Architecture	Ajuster le taux d'apprentissage, les neurones, le dropout et la régularisation L2.	Améliorer la généralisation.
Interprétabilité	Ajouter SHAP ou l'importance par permutation.	Identifier les facteurs de churn les plus déterminants.
Sauvegarde	Utiliser model.save("churn_model.keras") .	Suivre la recommandation Keras actuelle.

3.6 Conclusion pratique:

Le notebook implémente avec succès un pipeline pratique complet de prédiction du churn. Il part des données clients brutes, nettoie et prétraite les variables, entraîne un classifieur par réseau de neurones, évalue le modèle à l'aide de plusieurs métriques, visualise les performances, teste un client individuel et sauvegarde le modèle entraîné.

La performance obtenue est acceptable pour une référence de base : l'accuracy de test est de 78,46 % et la ROC-AUC d'environ 0,82. Néanmoins, la faiblesse pratique réside dans la détection des churners : le rappel du modèle pour la classe Churn est de 55,88 %, avec 165 faux négatifs dans l'ensemble de test. Par conséquent, la prochaine étape pratique devrait porter sur l'optimisation du seuil, le traitement du déséquilibre des classes, l'EarlyStopping et l'interprétabilité du modèle.

Annexe: Récapitulatif compact de toutes les étapes du notebook:

Table 3.6 – Récapitulatif compact des étapes pratiques du notebook

1. Importation	Environnement Python	Chargement pandas, numpy, sklearn, TensorFlow/Keras	Librairies prêtes
2. Chargement	URL GitHub	pd.read_csv	DataFrame df
3. Exploration	df	shape, columns, info, describe, value_counts, graphique	Compréhension initiale
4. Nettoyage	df brut	Conversion TotalCharges, suppression valeurs manquantes/ customerID, mapping Churn	df nettoyé
5. Séparation X/y	df nettoyé	Retrait de Churn pour X, extraction de Churn comme y	X : 19 variables, y : cible
6. Types de variables	X	select_dtypes	numeric_cols et categorical_cols

7. Découpage train/test	X et y	Découpage stratifié 80/20	Sous-ensembles d'entraînement et de test
8. Prétraitement	train/test bruts	StandardScaler + OneHotEncoder	Matrices numériques à 45 variables
9. Modèle	45 variables	Sequential Dense/ Dropout/Sigmoid	Réseau à 5 057 paramètres
10. Compilation	Modèle	Adam + binary_crossentropy + accuracy	Modèle compilé
11. Entraînement	X_train_processed	50 époques, batch 32, validation 20 %	Objet history
12. Évaluation	X_test_processed	model.evaluate	Perte et accuracy
13. Prédiction	Probabilités	Seuil 0,5	Classes 0/1
14. Matrice de confusion	y_test, y_pred	confusion_matrix	TN, FP, FN, TP
15. Rapport de classification	y_test, y_pred	precision, recall, F1-score	Diagnostic par classe
16. Visualisations	history et cm	courbes accuracy/loss + heatmap	Analyse graphique
17. Exemple individuel	Un client de test	transform + predict	Probabilité et décision
18. ROC	Probabilités prédites	roc_curve + auc	AUC ≈ 0,82
19. Sauvegarde	Modèle entraîné	model.save("churn_model.h5")	Fichier modèle sauvegardé
20. Niveau de risque	Probabilité	Règles de seuil 0,7/0,4	Risque Élevé/Moyen/ Faible

Conclusion Générale:

Ce rapport a permis d'articuler un cadre théorique et une mise en œuvre pratique autour d'un même objectif : la prédiction du churn client à l'aide du Machine Learning et des réseaux de neurones artificiels. La partie théorique a posé les fondements conceptuels — définitions, historique, typologie des apprentissages, algorithmes de classification, architecture des réseaux de neurones, fonctions d'activation, processus d'apprentissage par rétropropagation, ainsi que les métriques d'évaluation (matrice de confusion, précision, rappel, F1-score, courbe ROC et AUC). La partie pratique a directement mobilisé ce cadre conceptuel en implémentant, sur le jeu de données Telco Customer Churn, un réseau de neurones Dense/ Dropout/Sigmoid entraîné avec l'optimiseur Adam et évalué au moyen des mêmes métriques présentées au chapitre 1.

Les résultats pratiques confirment plusieurs constats théoriques. D'une part, la performance globale du modèle (accuracy de 78,46 % et ROC-AUC d'environ 0,82) illustre la capacité des réseaux de neurones à modéliser des relations non linéaires complexes entre les caractéristiques clients et le comportement de résiliation, comme souligné au chapitre 2. D'autre part, le rappel modéré obtenu pour la classe Churn (55,88 %) et l'écart croissant entre les courbes d'entraînement et de validation illustrent concrètement deux limites théoriques identifiées dans la partie 1 : la sensibilité des modèles au déséquilibre des classes et le risque de surapprentissage propre aux architectures de Deep Learning. De même, l'absence d'analyse d'explicabilité dans le notebook renvoie directement à la limite du « black-box » évoquée dans la discussion théorique sur les avantages et limites des réseaux de neurones.

Ainsi, la confrontation entre théorie et pratique met en évidence la cohérence du cadre conceptuel mobilisé, tout en révélant les arbitrages concrets que rencontre tout praticien :

choix du seuil de décision, traitement du déséquilibre des classes, compromis entre performance et interprétabilité. Les recommandations pratiques formulées à l'issue de l'analyse — EarlyStopping, rééquilibrage des classes, optimisation du seuil, ajout de techniques d'explicabilité — s'inscrivent directement dans le prolongement des bonnes pratiques identifiées dans la littérature théorique. Ce rapport démontre ainsi que la prédiction du churn client, loin de se limiter à un exercice technique, nécessite une articulation constante entre les fondements conceptuels du Machine Learning et les contraintes opérationnelles propres à chaque jeu de données, afin de transformer les données clients brutes en connaissance métier exploitable pour la rétention et la fidélisation.

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Les références suivantes couvrent l'ensemble du rapport (partie théorique et partie pratique). Pour la partie pratique, les sources techniques correspondent à la documentation officielle des librairies utilisées dans le notebook (pandas, scikit-learn, TensorFlow/Keras) et au jeu de données Telco Customer Churn.

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